Discrete Mathematics Cheatsheet

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1 Propositional Logic

1.1 Logical Connectives

1.1.1 Logical Operators

Symbol	Name	Meaning
$\neg p$	Negation	NOT p
$p \wedge q$	Conjunction	p AND q
$p \vee q$	Disjunction	p OR q
$p \to q$	Implication	IF p THEN q
$p \leftrightarrow q$	Biconditional	IFF
$p \oplus q$	XOR	Exactly one

1.1.2 Truth Table

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \to q$	$p \leftrightarrow q$	$p \oplus q$
T	T	F	T	T	T	T	F
T	F	F	F	T	F	F	T
F	T	T	F	T	T	F	T
F	F	T	F	F	T	T	F

1.2 Propositional Equivalences

1.2.1 Logical Equivalences

Law	Equivalence
Identity	$p \wedge T \equiv p, \quad p \vee F \equiv p$
Domination	$p \vee T \equiv T, \quad p \wedge F \equiv F$
Idempotent	$p \wedge p \equiv p, \quad p \vee p \equiv p$
Double Negation	$\neg\neg p \equiv p$
Commutative	$p \wedge q \equiv q \wedge p$ $p \vee q \equiv q \vee p$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$ $(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$
Negation	$p \vee \neg p \equiv T, \quad p \wedge \neg p \equiv F$
Absorption	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$

Law	Equivalence
Conditional	$p \to q \equiv \neg p \vee q$
Converse (of $p \to q$)	$q \to p$
Inverse (of $p \to q$)	$\neg p \to \neg q$
Contrapositive (equivalent)	$\neg q \to \neg p$
Negation of Conditional	$\neg(p \to q) \equiv p \wedge \neg q$

1.2.2 Logical Equivalences Involving Conditional and Biconditional Statements

Law	Equivalence
Biconditional Identity	$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
Biconditional (Expanded)	$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
Negation of Biconditional	$\neg(p \leftrightarrow q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

1.2.3 Translation of Logical Equivalences

English Phrase	Logical Form
not p	$\neg p$
p and q	$p \wedge q$
p or q	$p \vee q$
Either p or q, but not both	$p \oplus q$
If p, then q	$p \to q$
p only if q	$p \to q$
p if q	$q \to p$
p if and only if q	$p \leftrightarrow q$
p is sufficient for q	$p \to q$
p is necessary for q	$q \to p$
Neither p nor q	$\neg p \wedge \neg q$
Not both p and q	$\neg(p \wedge q)$
At least one of p, q	$p \vee q$
Both p and q are false	$\neg p \wedge \neg q$

1.3 Predicates and Quantifiers

Symbol	Name	Meaning
$\forall x\, P(x)$	Universal	$P(x)$ holds for every x
$\exists x\, P(x)$	Existential	$P(x)$ holds for at least one x
$\exists! x\, P(x)$	Unique existence	$P(x)$ holds for exactly one x

1.3.1 De Morgan's Laws for Quantifiers

Law	Equivalence
Universal Negation	$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$
Existential Negation	$\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$
Negated Universal	$\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)$
Negated Existential	$\neg(\exists x\, P(x)) \equiv \forall x\, \neg P(x)$

1.3.2 Multiple Quantifiers

Statement	Meaning
$\forall x \forall y\, P(x, y)$	P holds for every pair (x, y)
$\exists x \exists y\, P(x, y)$	P holds for at least one pair
$\forall x \exists y\, P(x, y)$	For every x, some y works
$\exists y \forall x\, P(x, y)$	One y works for every x

1.3.3 Properties of Quantifiers

Property	Equivalence
Universal Distribution	$\forall x(P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$
Existential Distribution	$\exists x(P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$
Universal Commutativity	$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$
Existential Commutativity	$\exists x \exists y P(x, y) \equiv \exists y \exists x P(x, y)$
Universal Implication	$\forall x(P(x) \to Q(x)) \equiv \neg\exists x(P(x) \wedge \neg Q(x))$
Existential Implication	$\exists x(P(x) \to Q(x)) \equiv \neg\forall x P(x) \vee \exists x Q(x)$

1.3.4 Quantifier Order

Property	Relationship
Same Quantifiers	$\forall x \forall y P \equiv \forall y \forall x P$ $\exists x \exists y P \equiv \exists y \exists x P$
Mixed Quantifiers	$\exists y \forall x P(x, y) \to \forall x \exists y P(x, y)$
Not Equivalent	$\forall x \exists y P(x, y) \not\equiv \exists y \forall x P(x, y)$

1.3.5 Translation into Predicate Logic

English	Logical Form
Every x has property P	$\forall x\, P(x)$
Some x has property P	$\exists x\, P(x)$
No x has property P	$\forall x\, \neg P(x)$
Not every x has property P	$\exists x\, \neg P(x)$
Everyone is P and Q	$\forall x\, (P(x) \wedge Q(x))$
Everyone is P or Q	$\forall x\, (P(x) \vee Q(x))$
Someone is P and Q	$\exists x\, (P(x) \wedge Q(x))$
Someone is P or Q	$\exists x\, (P(x) \vee Q(x))$
Every P is Q	$\forall x\, (P(x) \to Q(x))$
Some P is Q	$\exists x\, (P(x) \wedge Q(x))$
No P is Q	$\forall x\, (P(x) \to \neg Q(x))$
Some P is not Q	$\exists x\, (P(x) \wedge \neg Q(x))$

1.4 Rules of Inference

Rule	Form
Modus Ponens (MP)	$\dfrac{p \quad p \to q}{\therefore q}$
Modus Tollens (MT)	$\dfrac{p \to q \quad \neg q}{\therefore \neg p}$
Hypothetical Syllogism (HS)	$\dfrac{p \to q \quad q \to r}{\therefore p \to r}$
Disjunctive Syllogism (DS)	$\dfrac{p \vee q \quad \neg p}{\therefore q}$
Addition	$\dfrac{p}{\therefore p \vee q}$
Simplification	$\dfrac{p \wedge q}{\therefore p}$
Conjunction	$\dfrac{p \quad q}{\therefore p \wedge q}$
Resolution	$\dfrac{p \vee q \quad \neg p \vee r}{\therefore q \vee r}$
Universal Instantiation (UI)	$\dfrac{\forall x\, P(x)}{\therefore P(c)}$
Existential Instantiation (EI; fresh c)	$\dfrac{\exists x\, P(x)}{\therefore P(c),\ c \text{ fresh}}$
Universal Generalization (UG)	$\dfrac{P(c),\ c \text{ arbitrary}}{\therefore \forall x\, P(x)}$
Existential Generalization (EG)	$\dfrac{P(c)}{\therefore \exists x\, P(x)}$

1.5 Introduction to Proofs

1.5.1 Choosing a Proof Method

Methods of Proving
Can we derive q directly from p? **Direct Proof**
Is $\neg q \to \neg p$ easier to establish? **Proof by Contrapositive**
Does assuming the statement is false lead to a contradiction? **Proof by Contradiction**
Can we divide the problem into exhaustive cases? **Proof by Cases**
Can we prove both $p \to q$ and $q \to p$? **Proof of Equivalence**
Can we find an object satisfying the required property? **Existence Proof**
Can we start from the conclusion and work backward? **Backward Reasoning**

1.5.2 Mathematical Induction

To prove $P(n)$ for all $n \geq n_0$:

1. **Base Case:** Verify $P(n_0)$.
2. **Inductive Hypothesis:** Assume $P(k)$ is true for arbitrary $k \geq n_0$.
3. **Inductive Step:** Use $P(k)$ to prove $P(k+1)$.
4. **Conclusion:** By induction, $P(n)$ holds for all $n \geq n_0$.

2 Proof

2.1 Direct Proof

1. If n is odd, n^2 is odd.
Assume that n is odd.
Let $n = 2k+1$, $k \in \mathbb{Z}$.

$$n^2 = (2k+1)^2 = 2(2k^2+2k)+1.$$

Hence n^2 is odd. □
2. If m, n are squares, mn is a square.
Assume that m and n are both squares.
Let $m = a^2, n = b^2$, $a, b \in \mathbb{Z}$.

$$mn = a^2b^2 = (ab)^2.$$

Hence mn is a perfect square. □

2.2 Proof by Contraposition

If $3n+2$ is odd, n is odd.
Contrapositive:
Assume that n is even, $3n+2$ is even.
Let $n = 2k$:

$$3n+2 = 6k+2 = 2(3k+1).$$

Since the negation of conclusion implies the negation of hypothesis, the original conditional statement is true. □
2. If $n = ab$, then $a \le \sqrt{n}$ or $b \le \sqrt{n}$.
Contrapositive:
Assume that $a > \sqrt{n}$ and $b > \sqrt{n}$, then $ab \neq n$.
Since $a, b > 0$,

$$ab > \sqrt{n} \cdot \sqrt{n} = n.$$

Thus $ab \neq n$.
Since the negation of conclusion implies the negation of hypothesis, the original conditional statement is true. □

2.3 Proof by Contradiction

1. $\sqrt{2}$ is irrational.
Assume that $\sqrt{2}$ is rational.
$\sqrt{2} = a/b$, $a, b \in \mathbb{Z}$, $\quad b \neq 0$, $\quad a, b$ coprime

$$a^2 = 2b^2 \Rightarrow a = 2k$$

$$4k^2 = 2b^2 \Rightarrow b^2 = 2k^2.$$

Both a, b are even, they share a common factor of 2.
It contradicts the initial assumption that a, b coprime, so the original statement is true. □
2. If $3n+2$ is odd, n is odd.
Assume that the statement is false. Then we have $3n+2$ is odd and n is even. Let $n = 2k$:

$$3n+2 = 2(3k+1),$$

Hence $3n+2$ is even.
It contradicts the initial assumption that $3n+2$ is odd, so the original statement is true. □

2.4 Proof of Equivalence

Prove: P_1: n even; P_2: $n-1$ odd; P_3: n^2 even.
Show $P_1 \Rightarrow P_2 \Rightarrow P_3 \Rightarrow P_1$.
$P_1 \Rightarrow P_2$**:** $n = 2k \Rightarrow n-1 = 2(k-1)+1$ (odd).
$P_2 \Rightarrow P_3$**:** $n-1 = 2k+1 \Rightarrow n = 2(k+1)$.

$$n^2 = 2[2(k+1)^2] \quad \text{(even)}.$$

$P_3 \Rightarrow P_1$**:** Contrapositive: $n = 2k+1$ gives

$$n^2 = 2(2k^2+2k)+1 \quad \text{(odd)}.$$

Hence all three are equivalent. □

2.5 Non-Constructive Proof

Exist irrational x, y with rational x^y.
Let $a = (\sqrt{2})^{\sqrt{2}}$.
Case 1: If a is rational, choose $x = y = \sqrt{2}$. Then $x^y = (\sqrt{2})^{\sqrt{2}}$ is rational.
Case 2: If a is not rational, choose $x = a$, $y = \sqrt{2}$:

$$x^y = ((\sqrt{2})^{\sqrt{2}})^{\sqrt{2}} = 2.$$

, which is rational.
Both cases establish existence. □

2.6 Backward Reasoning

AM–GM: For $a, b > 0$, prove $\frac{a+b}{2} \ge \sqrt{ab}$.
Work backward:

$$\begin{aligned}\frac{a+b}{2} \ge \sqrt{ab} &\iff a+b \ge 2\sqrt{ab} \\ &\iff a+b-2\sqrt{ab} \ge 0 \\ &\iff (\sqrt{a}-\sqrt{b})^2 \ge 0.\end{aligned}$$

Last statement is always true. Equality iff $a = b$. □

3 Mathematical Induction

3.1 Sum of Odd Integers

Prove: $1+3+\cdots+(2n-1) = n^2$, $n \ge 1$.
Base step: $n = 1$: $1 = 1^2$.
Induction Hypothesis:
Assume that the statement holds for $n = k, k \in \mathbb{Z}^+$.
$1+3+\cdots+(2k-1) = k^2$.
Inductive step:

$$\begin{aligned}1+3+\cdots+(2k+1) &= k^2+(2k+1) \\ &= (k+1)^2.\end{aligned}$$

so that $P(k+1)$ is true if $P(k)$ is true.
Thus, the statement holds for all $n \in \mathbb{Z}^+$. □

3.2 Inequality: $n < 2^n$

Prove: $n < 2^n$, $n \ge 1$.
Base step: $n = 1$: $1 < 2$.
Thus the statement holds for $n = 1$
Inductive Hypothesis:
Assume that the statement holds for $n = k, k \in \mathbb{Z}^+, k \ge 1$
$k < 2^k$.
Inductive step: Since $k \ge 1$,

$$k+1 \le 2k < 2 \cdot 2^k = 2^{k+1}.$$

so that $P(k+1)$ is true if $P(k)$ is true.
Thus, $n < 2^n$ is true for all $n \in \mathbb{Z}^+$. □

3.3 Divisibility by 3

Prove: $3 \mid (n^3 - n)$, $n \ge 1$.
Base step: $n = 1$: $1^3 - 1 = 0$.
Inductive Hypothesis:
Assume that the statement holds for $n = k, k \in \mathbb{Z}^+$ $3 \mid (k^3 - k)$.
Inductive step:

$$\begin{aligned}(k+1)^3-(k+1) &= k^3+3k^2+2k \\ &= (k^3-k)+3k(k+1).\end{aligned}$$

By the inductive step, we conclude that both terms are divisible by 3. Thus, $3 \mid (n^3-n)$ for all $n \in \mathbb{Z}^+$.
□

3.4 Sum of Squares

Prove:

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}.$$

Base step: $n = 1$: $1 = \frac{1(2)(3)}{6}$.
Thus the statement holds for $n = 1$.
Inductive Hypothesis:
Assume the statements holds for $n = k, k \in \mathbb{Z}^+$.

$$\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}.$$

Inductive step:

$$\begin{aligned}\sum_{i=1}^{k+1} i^2 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= \frac{(k+1)[k(2k+1)+6(k+1)]}{6} \\ &= \frac{(k+1)(2k^2+7k+6)}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6}.\end{aligned}$$

This is the formula for $n = k+1$.
By mathematical induction, the formula is true for all $n \in \mathbb{Z}^+$. □

3.5 Factorial Inequality

Prove: $2^n < n!$, $n \ge 4$.
Base step: $n = 4$: $16 < 24$.
Thus the statement holds for $n = 4$ **Inductive Hypothesis:** Assume the statement holds for $n = k, k \in \mathbb{Z}^+, k \ge 4$.
$2^k < k!$
Inductive step: Since $k+1 > 2$,

$$2^{k+1} = 2 \cdot 2^k < 2k! < (k+1)k! = (k+1)!.$$

Thus, $P(k+1)$ is true if $P(k)$ is true, $2^n < n!$ for all $n \in \mathbb{Z}^+$ where $n \ge 4$. □

3.6 Geometric Sum

Prove:

$$1+2+2^2+\cdots+2^n = 2^{n+1}-1, \quad n \ge 0.$$

Base step: $n = 0$: $1 = 2^1 - 1$. Thus, the statement holds for $n = 0$
Inductive Hypothesis:
Assume that the statement holds for $n = k, k \in \mathbb{Z}_{\ge 0}$
$\sum_{i=0}^{k} 2^i = 2^{k+1} - 1$.
Inductive Step:

$$\begin{aligned}\sum_{i=0}^{k+1} 2^i &= (2^{k+1}-1)+2^{k+1} \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{k+2} - 1.\end{aligned}$$

By mathematical induction, $P(n)$ is true for all $n \ge 0$. □

3.7 Divisibility by 57

Prove: $57 \mid (7^{n+2} + 8^{2n+1})$, $n \ge 0$.
Base step: $n = 0$:

$$7^2 + 8^1 = 49 + 8 = 57.$$

Inductive Hypothesis:
Assume the statement holds for $n = k, k \in \mathbb{Z}, k \ge 0$
$57 \mid (7^{k+2} + 8^{2k+1})$.
Inductive step:

$$\begin{aligned}&7^{k+3} + 8^{2k+3} \\ &= 7 \cdot 7^{k+2} + 64 \cdot 8^{2k+1} \\ &= 7(7^{k+2} + 8^{2k+1}) + 57 \cdot 8^{2k+1}.\end{aligned}$$

By the inductive step, we conclude that first term is divisible by 57, and the second term is also divisible by 57. Thus, $7^{k+2} + 8^{2k+1}$ is divisible by 57 for all $n \in \mathbb{Z}$ where $n \ge 0$
Hence true for all $n \ge 0$. □